NOTES:

1. REFER TO CONTROL PHILOSOPHY FOR COMPREHENSIVE LIST OF PROCESS INTERLOCKS AND CONTROL FUNCTIONS.
2. STAGE 2.
3. TIE IN POINT LOCATED ON F17-BX-410.
4. SIGNAL FROM FLOTATION THICKENER LCP TO MCC/DCS.
5. FLOCCULANT IS BEING DISTRIBUTED WITHIN VENDOR PACKAGE.
6. THERE ARE 3 OFF SPARE NOZZLES (DN100, DN200 & DN60 AVAILABLE).
7. FLANGED ASME B16.5 CLASS 150.

HIGH PRESSURE PROCESS WATER PUMPS
2000-F17-PID-PR-10002
HP PROCESS WATER
17-0174-PW-PE16-160

MILL WATER PUMPS
2000-F24-PID-PR-10001
PROCESS WATER
17-0360-PW-PE8-225
17-0361-PW-SS1-200
200VBFD1S2 F17-10837
200VBFD1S1 F17-10838
50VB03S1 F17-10841
17-0925-PW-SS3-50
TO GRADE
150VBFD1S1 FC
200x150
200VBFD1S1 F17-10839
200VBFD1S1 F17-10840
200VCD1S1 F17-10842
300x200

FLOCCULANT DOSING PUMPS
2000-F72-PID-PR-10004
FLOCCULANT
17-0371-FL-PE8-90
17-0372-FL-MS1-80
80VBD1C1 F17-10843
80VBD1C1 F17-10844
25VB03S1 F17-10845
17-0275-FL-MS1-25
TO GRADE

FLOTATION AREA SUMP PUMP 2
2000-F16-PID-PR-10003
SPILLAGE
16-1208-FD-PE10-160

FLOTATION AREA SUMP PUMP 3
2000-F16-PID-PR-10004
SPILLAGE
16-1214-FD-PE10-160

CLEANER AREA SUMP PUMP
2000-F18-PID-PR-10001
SPILLAGE
18-1216-FD-PE10-110

FLOCCULANT AREA SUMP PUMP
2000-F72-PID-PR-10004
SPILLAGE
72-1076-FL-PE8-90

FLOTATION TAILINGS THICKENER UNDERFLOW PUMP
2000-F17-PID-PR-10002
FLOTATION TAILINGS THICKENER UNDERFLOW
17-0383-CON-PE10-500

MILL WATER PUMPS
2000-F24-PID-PR-10001
PROCESS WATER
17-0187-PW-PE8-500
500VBFD1L2 F17-10846

FLOTATION CELLS DRAIN PUMP
2000-F16-PID-PR-10003
SCAVENGER TAILS
16-1290-FS-PE10-280

FLOTATION AREA SUMP PUMP 1
2000-F16-PID-PR-10001
SPILLAGE
16-1206-FD-PE10-160

RAW WATER PUMPS
2000-F24-PID-PR-10002
RAW WATER
17-0300-BW-PE8-63
17-0975-BW-SS3-50
50VC01S1 F17-10847
17-1424-BW-SS3-25
50x25
25VB03S1 F17-14149
25VB03S1 F17-14109
25VB03S1 F17-10848
25VB03S1 F17-10849
SP HS-056
SP HS-056
SP MC-001
SP HS-025
F17-XM-572
FLOTATION TAILINGS THICKENER
¼" MANTA SUB BSP FEMALE

FLOTATION DISCHARGE BOX
2000-F16-PID-PR-10003
SCAVENGER TAILS
16-1248-FS-RL1-1200
1.5°
ID25mm
ID25mm
DN1200
DN1200
F17-SA-416
FLOTATION TAILINGS THICKENER FEED SAMPLER
VENDOR PACKAGE 2000-ME074-J02-10007
DN50
17-1742-FS-RL1-1200
17-1743-FS-RL1-80
80VKGD1L1 F17-10851
TO GRADE
SP HS-105
SP HS-105
1200VKGD1L2 FL
F17-BX-410
FLOTATION TAILINGS THICKENER RECIRCULATION FEED BOX
17-0497-FS-RL1-1200
TI 17141
PI 17140
LI 17142
F17-HU-422
FLOTATION TAILINGS THICKENER FEED BOX VALVE HYDRAULIC POWER UNIT
F17-XM-474 F17-XM-471
F17-XM-613 F17-XM-472
F17-XM-614 F17-XM-473
FLOTATION TAILINGS THICKENER OVERFLOW LAUNDER DART VALVES (6 OFF)
THICKENER FEED STANDPIPE NOTE 6
F17-PP-459
FLOTATION TAILINGS P-DUC PUMP
F17-TH-401
FLOTATION TAILINGS THICKENER

PE10
17-0368-FD-PE10-225
RL1
17-0367-FD-RL1-200
50VB03S1 F17-10853
SP MC-002
LAH 17081
200VKGD1L1 F17-10854
50VB03S1 F17-10852
SP MC-002
LSH 17081
200x100
NOTE 7
F17-PS-427
FLOTATION TAILINGS THICKENER AREA SUMP PUMP 1

16-0496-FS-PE10-63
LIC 17004 H L LL
LT 17004
LI 17004
DETAIL 3 HS 17005
XV 17005
80VKGD1L1 FL
F16-HP-456
FLOTATION TAILS SAMPLE HOPPER
16-3072-FS-PE10-110
TO GRADE
SP MC-001
25VB03S1 F16-10855
ZAH 17144
PAH 17143
(6 OFF)
BURST
XA 17008
XS 17008
SP HS-074
VSD
16-0660-FS-RL1-80
16-2140-FS-RL1-80
80x65
80x65
SP HS-074
F16-PP-634
FINAL TAILS SAMPLE PUMP
DETAIL 4 HS 17006
XV 17006
50VKGD1L1 FO
RL1
SS3
16-1515-FS-SS3-50
TO GRADE
25VB03S1 F17-10856
DETAIL 4 HS 17007
XV 17007
50VKGD1L1 FO
RL1
SS3
16-1448-FS-SS3-50
TO GRADE
SP MC-001
RL1
PE10
90x63

FIT 17001
FE 17001
DETAIL 1
FIC 17001 H L
FCV 17001
f(x)
DILUTION FACTOR
PE8
SS1
M
FIT 17003
FE 17003
M
PE16
SS3
17-0130-PW-SS3-50
17-0267-PW-PE16-63
FI 17003
F17-XM-413
FLOTATION TAILINGS THICKENER FLOCCULANT INLINE MIXER
SP HS-100
25VB03S1 F17-10821
50VB03S1 F17-10822
SP MC-001
SP MC-002
SP @ BRIDGE OF THICKENER F17-TH-401
17-0373-FL-SS1-300
CCTV F17-CAM-001
SP SC-034
NOTE 5
VSD
DN150
TAH 17039
H
M
G
PI 17002
PI 17002
PT 17002
50VBE1S1 F17-10823
F17-HC-461
F17-HD-461
FLOTATION TAILINGS THICKENER BRIDGE MONORAIL AND TROLLEY

17-0982-PW-PE16-160
17-0983-PW-PE16-160
PE16
SS2
SS3
17-0976-PW-SS2-100
17-1487-PW-SS3-50
17-0984-PW-SS2-100
17-1486-PW-SS3-50
17-0985-PW-SS2-100
17-1485-PW-SS3-50
17-0986-PW-SS2-100
17-1484-PW-SS3-50
17-0987-PW-SS2-100
17-1483-PW-SS3-50
17-0988-PW-SS2-100
17-1482-PW-SS3-50
100VBFE1S1 F17-10824
50VB03S1 F17-10825
100VBFE1S1 F17-10826
50VB03S1 F17-10827
100VBFE1S1 F17-10828
50VB03S1 F17-10829
100VBFE1S1 F17-10830
50VB03S1 F17-10831
100VBFE1S1 F17-10832
50VB03S1 F17-10833
100VBFE1S1 F17-10834
50VB03S1 F17-10835
SP MC-002
F24-XM-451
F24-XM-452
F24-XM-453
F24-XM-454
F24-XM-455
F24-XM-456
FLOTATION TAILINGS THICKENER WATER MONITORS (6 OFF) NOTE 7

ZS 17010
TURBIDITY
AI 17019 HH H
LIT 17012
LE 17012
M
BED LEVEL METER MANTA SUB
RAKE HEIGHT LO
ZSL 17015
RAKE HEIGHT HI
ZSH 17014
RAKE UNDERSPEED SWITCH
SSL 17016
GEARBOX LEVEL SWITCH LOW
LSL 17018
GEARBOX LEVEL SWITCH LOW
LSL 17017
RAKE HEIGHT POSITION
ZIT 17013
ZE 17013
F17-HU-406
FLOTATION TAILINGS THICKENER RAKE DRIVE HPU
F17-HU-414
FLOTATION TAILINGS THICKENER RAKE LIFT HPU
AIT 17019
AE 17019
VENDOR PACKAGE 2000-ME0015-E02-50001 TO 50006

MANTA SUB START/STOP HS 17038
MANTA SUB RUNNING XI 17038A
MANTA SUB FAULT XA 17038B
MANTA SENSOR FAILED TO RETURN HOME ALARM XA 17038
NOTE 4
XA 17060 EMERGENCY STOP
ZAH 17014 MAX. RAKE LEVEL
ZAL 17015 MIN. RAKE LEVEL
XI 17066 24VDC POWER HEALTHY
SV 17033 RAKE DRIVE FORWARD SOLENOID RELAY
SV 17034 RAKE DRIVE REVERSE SOLENOID RELAY
SV 17048 RAKE LIFT RAISE SOLENOID RELAY
SV 17047 RAKE LIFT LOWER SOLENOID RELAY
XC 17076 RAKE DRIVE RUNNING FEEDBACK
XC 17057 LOCAL MODE ENABLE FEEDBACK
NI 17030 HH H TORQUE INDICATION
XC 17079 85% TORQUE FROM DCS TO LCP
XC 17083 50% TORQUE FROM DCS TO LCP
XC 17080 40% TORQUE FROM DCS TO LCP
XC 17084 30% TORQUE FROM DCS TO LCP
HS 17065A RAKE LIFT LOCAL RAISE REQUEST
HS 17065B RAKE LIFT LOCAL LOWER REQUEST
HS 17063 REVERSE JOG LOCAL START REQUEST
XC 17082 ALARM ACKNOWLEDGE FROM DCS TO LCP
XC 17062 REVERSE JOG ACKNOWLEDGE FROM DCS TO LCP
HS 17061 RAKE DRIVE FORWARD LOCAL START REQUEST
HS 17064 RAKE DRIVE LOCAL STOP REQUEST
PDAH 17036 RAKE DRIVE OIL TANK COOLING FAN OUTLET FILTER DIFF PRESSURE (WORKING)
PDAH 17028 RAKE DRIVE OIL TANK PUMP OUTLET FILTER DIFF PRESSURE
PDAH 17037 RAKE DRIVE OIL TANK COOLING FAN OUTLET FILTER DIFF PRESSURE (STANDBY)
PDAH 17031 RAKE DRIVE OIL TANK OUTLET OIL DIFF PRESSURE
ZI 17013 RAKE LIFT POSITION
TAH 17022 RAKE DRIVE OIL TANK TEMPERATURE HIGH
TAHH 17024 RAKE DRIVE OIL TANK TEMPERATURE HIGH
LALL 17023 RAKE DRIVE OIL TANK LEVEL LOW LOW
LAL 17021 RAKE DRIVE OIL TANK LEVEL LOW
LAL 17041 RAKE LIFT OIL TANK LEVEL LOW
TAH 17042 RAKE LIFT OIL TANK TEMPERATURE HIGH
LALL 17043 RAKE LIFT OIL TANK LEVEL LOW LOW
TAHH 17044 RAKE LIFT OIL TANK TEMPERATURE HIGH HIGH
PDAH 17051 RAKE LIFT OIL TANK PUMP OUTLET FILTER DIFF PRESSURE
LAL 17017 GEAR REDUCER OIL LEVEL LOW
LAL 17018 GEAR REDUCER OIL LEVEL LOW
SAL 17016 RAKE ZERO SPEED
ZAO 17026 F17-PO-408 THICKENER RAKE DRIVE PUMP SUCTION VALVE POSITION SWITCH OPENED

F17-VL-458
FLOTATION TAILINGS THICKENER OVERFLOW LAUNDER
F17-BX-464
FLOTATION TAILINGS THICKENER OVERFLOW BOX
F17-VL-490
FLOTATION TAILINGS THICKENER OVERFLOW PIPE LAUNDER 1
F17-VL-491
FLOTATION TAILINGS THICKENER OVERFLOW PIPE LAUNDER 2
SP SC-034
SP SC-034
SP SC-034
BED MASS METER
PE 17011
BED PRESSURE
PIC 17011 HH H L
DN50
50VB03S1 F17-10857
50VB03S1 F17-10858
SP MC-002
SP MC-002
SP MC-002
MECH
PROCESS WATER TANK 2000-F24-PID-PR-10001
FLOTATION TAILINGS THICKENER OVERFLOW
MECH
PROCESS WATER TANK 2000-F24-PID-PR-10001
FLOTATION TAILINGS THICKENER OVERFLOW
FIC17121
2000-F17-PID-PR-10002
SIGNAL
FIC17003
2000-F72-PID-PR-10004
SIGNAL

F17-TH-401
FLOTATION TAILINGS THICKENER
17-0498-CON-RL1-300
F17-10863
300VKGD1L1
TO GRADE
DN50
50VB03S1 F17-10859
SP MC-002
1:100
17-0364-CON-RL1-600
FLOTATION TAILINGS THICKENER UNDERFLOW PUMP 1
2000-F17-PID-PR-10002
FLOTATION TAILINGS THICKENER UNDERFLOW
DN50
50VB03S1 F17-10860
SP MC-002
1:100
17-0374-CON-RL1-600
FLOTATION TAILINGS THICKENER UNDERFLOW PUMP 1
2000-F17-PID-PR-10002
FLOTATION TAILINGS THICKENER UNDERFLOW
DN50
50VB03S1 F17-10861
SP MC-002
1:100
17-0365-CON-RL1-600
FLOTATION TAILINGS THICKENER UNDERFLOW PUMP 2
2000-F17-PID-PR-10002
FLOTATION TAILINGS THICKENER UNDERFLOW
DN50
50VB03S1 F17-10862
SP MC-002
1:100
17-0377-CON-RL1-600
FLOTATION TAILINGS THICKENER UNDERFLOW PUMP 2
2000-F17-PID-PR-10002
FLOTATION TAILINGS THICKENER UNDERFLOW
16-0661-FS-PE10-63
FLOTATION PARTICLE SIZE ANALYSER
2000-F16-PID-PR-10005
FINAL TAILS SAMPLE